Lab1

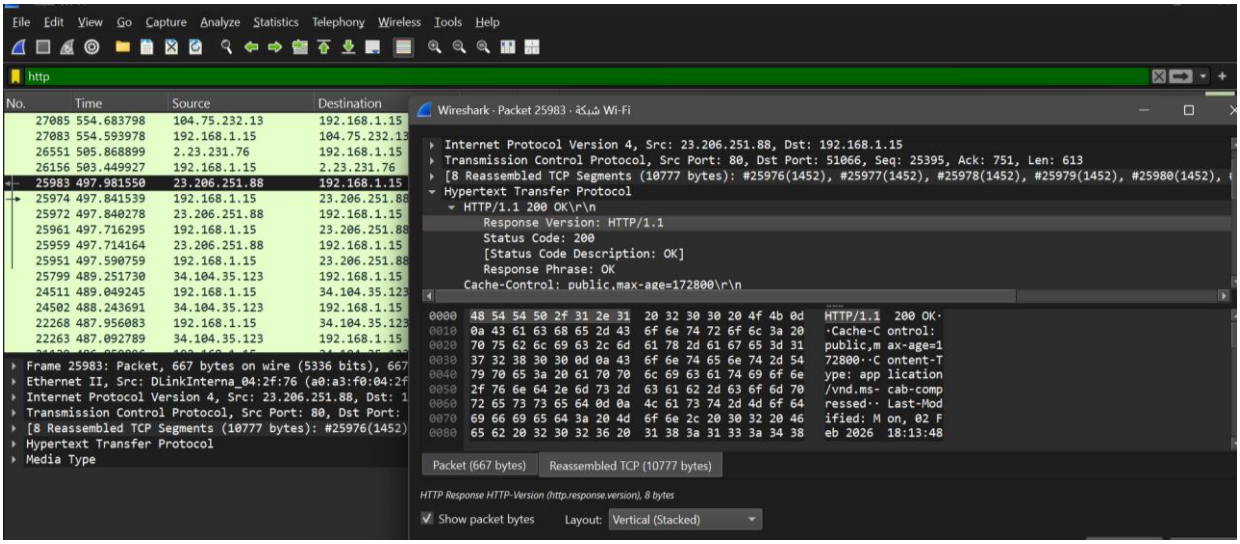
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Introduction

This lab aims to analyze common internet protocols using Wireshark in order to understand how data is transmitted over a network. Real network traffic was captured and examined to study the behavior of HTTP, HTTPS, TCP, and UDP protocols.

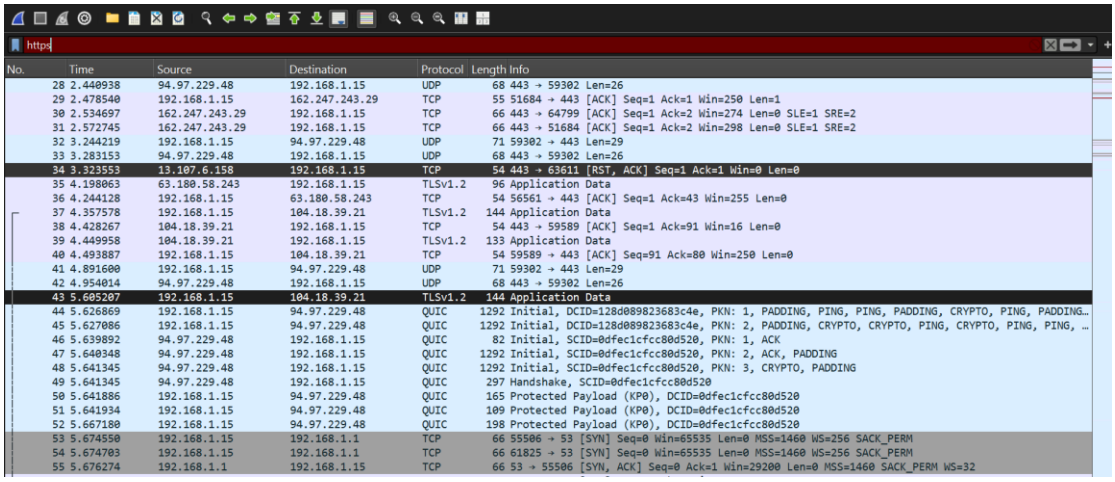
HTTP (Hypertext Transfer Protocol)

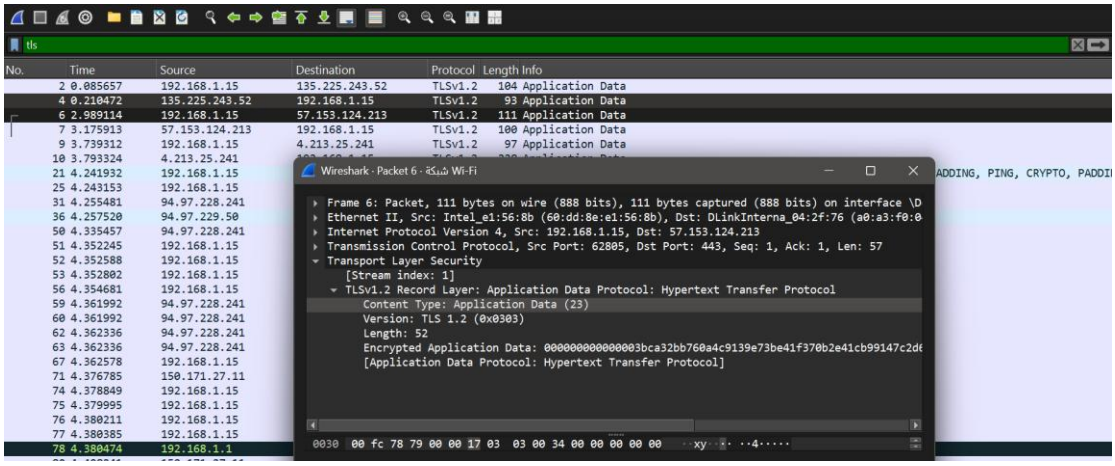
- Application-layer protocol used for transferring web pages.
- Works over **TCP**.
- Uses **port 80**.
- Data is transmitted in plain text (not encrypted).
- Common methods include **GET** and **POST**.
- Easy to analyze in Wireshark because the content is visible.



HTTPS (Hypertext Transfer Protocol Secure)

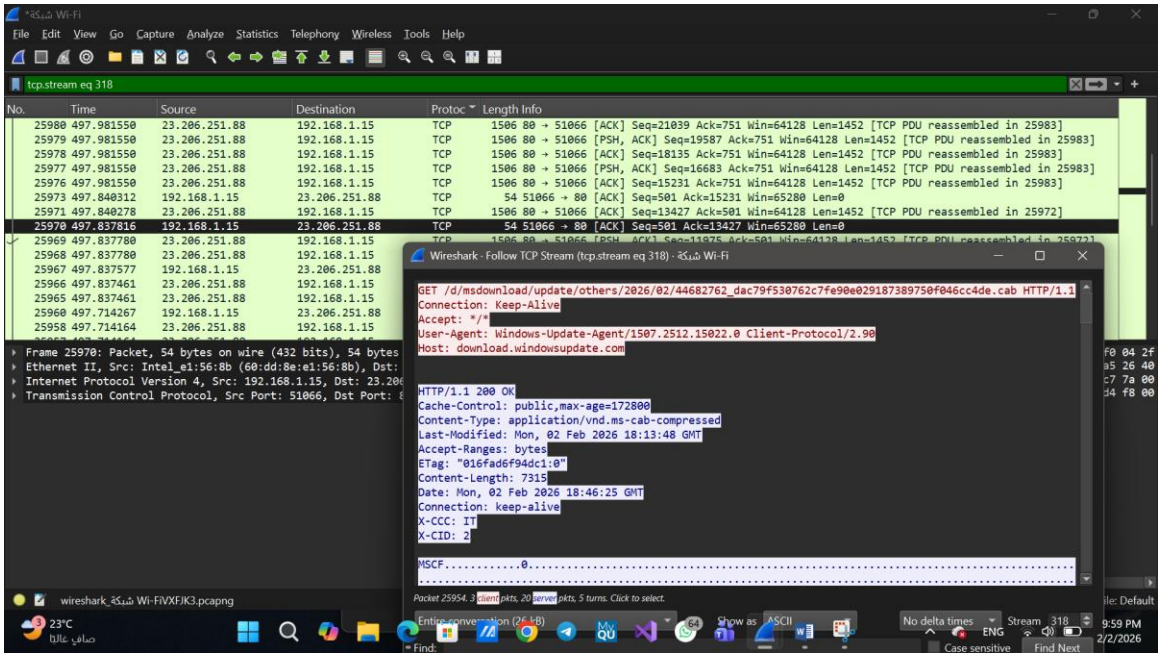
- Secure version of HTTP.
- Uses **TLS/SSL encryption** to protect data.
- Works over **TCP**.
- Uses **port 443**.
- Data is encrypted, so HTTP content cannot be read in Wireshark.
- Appears as **TLS packets** instead of HTTP packets.





TCP (Transmission Control Protocol)

- Connection-oriented and reliable protocol.
- Uses the **three-way handshake (SYN, SYN-ACK, ACK)** to establish connections.
- Ensures data integrity and correct ordering using sequence numbers.
- Retransmits lost packets.
- Slower than UDP due to reliability mechanisms.
- Commnly used by HTTP and HTTPS.



UDP (User Datagram Protocol)

- Connectionless and unreliable protocol.
- Does not use handshakes or acknowledgments.
- Faster than TCP due to minimal overhead.
- Does not guarantee data delivery or ordering.
- Uses a simple header.
- Commonly used for streaming, online gaming, and VoIP.

